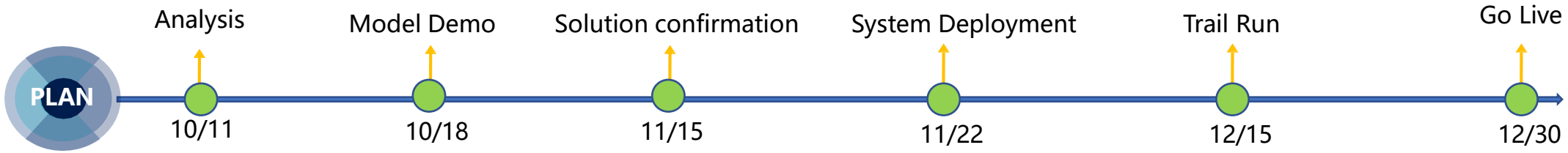
# DOE——Milestone

OT: 鄧雲濤　AT:范英秀/張文黃




| No | Execution Plan | Task | Current Status & Issues | Owner | Due Date | Status |
|---|---|---|---|---|---|---|
| 1 | Requirement Analysis | User Requirements Analysis | - Based on reflow oven temperature profile log data and product characteristics, predict temperature settings for 10 or 12 heating zones of the oven for the first trial run of a new product. | 范英秀/張文黃 | 10/08 | Closed |
| | | Data Collection & Preparation | - Collect oven settings, PCB characteristics data (Mr.Luong), 10/07, Done<br>- Collect material features (Ms.Que), 10/08, Done<br>- Collect Solder Paste data (Ms.Thom), 10/10, Done | 張文黃 | 10/11 | Closed |
| 2 | First Demo | First Model Demo | - Preprocess the data (CSV files) for model training<br>- Train a prediction model using 100 log data entries with the Gaussian Process Regression (GPR) algorithm. | 范英秀 | 10/18 | Closed |
| 3 | Solution confirmation | | | 鄧雲濤 | 11/15 | Closed |
| 4 | System Deployment | | - Confirm the models application method, 11/17, Done<br>- Build the API that connects the backend and frontend, 11/20, Done<br>- Deploy and launch, 11/22, Done | 范英秀/張文黃 | 11/22 | Closed |
| 5 | Trail Run | | - 1st week (11/24~11/26) => We havent received any user feedback about issues or anomalies that need improvement.<br>- 2nd week (12/2~12/6) The user plans to provide feedback on December 8.<br>- 3rd week (12/9) user feedback: Feasible and reduces testing time by 71%. (tested in 1 model PCB) | 范英秀/張文黃 | 12/15 | Closed |
| 6 | Go Live | | - 4th and 5th week (12/14 – 12/26) tested on 6 PCB models to evaluate overall performance and feasible => reduce testing time: 79% | 鄧雲濤 | 12/30 | Closed |

# DOE——Model demo

Link: https://fiisw-cns.myfiinet.com/genai-system/

## INPUT

### PCB physical parameters and conveyor speed

1. BOT: Number of bottom-side components
2. TOP: Number of top-side components
3. Panel: Number of panels on pcb
4. Length: PCB length
5. Width: PCB width
6. Height: PCB height
7. PCB has fixture? 1 – true; 0 – false
8. PCB has iron frame? 1 – true; 0 – false
9. Speed: conveyor speed

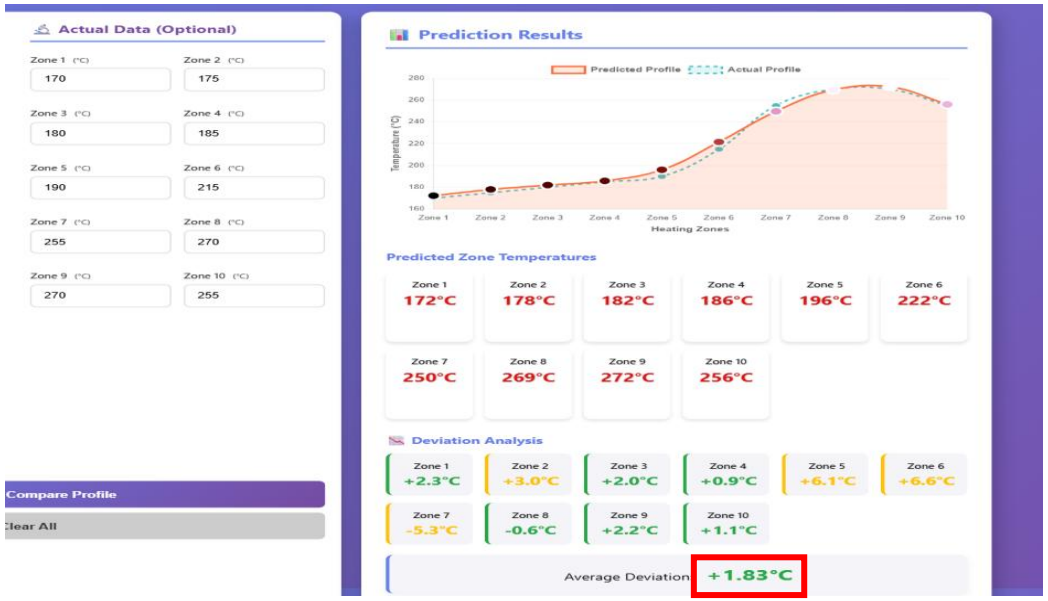
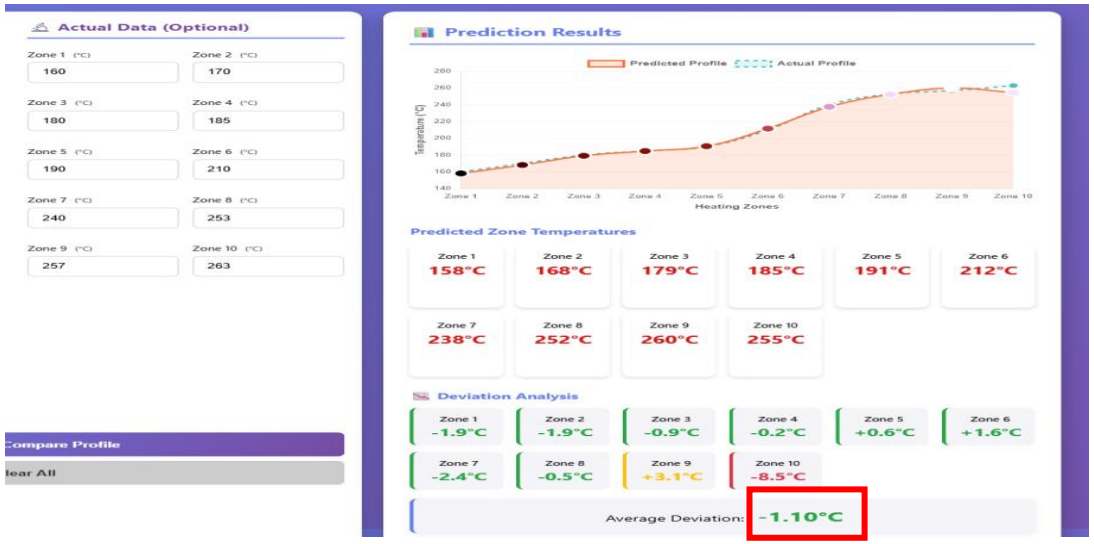
| 零件類數 | | Panel | Size(Panel) | | | 是否帶載具 | 是否帶鐵框 |
|---|---|---|---|---|---|---|---|
| BOT | TOP | | L (mm) | W (mm) | H(mm) | | |
| 197 | 153 | 12 | 161 | 183.6 | 1.573 | 是 | 是 |
| 228 | 184 | 10 | 187.8 | 159 | 1.6 | 否 | 是 |
| 216 | 199 | 10 | 187.2 | 159 | 1.6 | 否 | 是 |
| 203 | 187 | 10 | 187.2 | 159 | 1.6 | 否 | 是 |
| 258 | 222 | 12 | 184 | 137.6 | 1.6 | 是 | 是 |
| 385 | 223 | 4 | 160.4 | 182 | 1.2 | 是 | 是 |
| 27 | 124 | 4 | 158 | 174 | 1.2 | 是 | 否 |

## MODEL

## OUTPUT

### Heat setting for 10 zones

```
=== Input new product spec ===
BOT: 1478
TOP: 380
panel: 1
length: 168
width: 171.86
height: 2.31
fixture?: 1
iron frame?: 1
Speed: 90

=== 🔥 Profile predicted ===
Zone  1: 165.37 °C
Zone  2: 169.49 °C
Zone  3: 176.00 °C
Zone  4: 179.32 °C
Zone  5: 187.94 °C
Zone  6: 212.32 °C
Zone  7: 241.24 °C
Zone  8: 266.71 °C
Zone  9: 269.36 °C
Zone 10: 249.19 °C
```

Test results